

The Lagrange Top: Cubic Nutation, Two-Phase Reconstruction, and Exact Closure

Source-local theorem reconstruction

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Abstract

For the positive-inertia Lagrange symmetric heavy top we close the regular reduction–reconstruction problem. The inclination is inverted by one Jacobi function, both cyclic phases are complete third-kind elliptic integrals, and the full configuration closes exactly when two normalized phase increments are rational. Pole and repeated-root degenerations are kept outside singular Euler formulas. Exact computation certifies conventions, not the continuum proof.

1 Frozen convention and reduced cubic

Let $A = I_1 = I_2 > 0$, $C = I_3 > 0$, and $\gamma > 0$. In regular z - y - z Euler angles put $u = \cos \theta$, $L = p_\phi$, and $G = p_\psi$. Our convention is

$$H = \frac{p_\theta^2}{2A} + \frac{(L - Gu)^2}{2A(1 - u^2)} + \frac{G^2}{2C} + \gamma u. \quad (1)$$

The two momenta and $E = H$ are conserved. Since $p_\theta = A\dot{\theta}$, energy gives the reduced cubic owner

$$A^2 \dot{u}^2 = P(u) := 2A \left(E - \frac{G^2}{2C} - \gamma u \right) (1 - u^2) - (L - Gu)^2. \quad (2)$$

In particular

$$P(1) = -(L - G)^2, \quad P(-1) = -(L + G)^2. \quad (3)$$

The smooth Hamiltonian flow is complete: its potential is bounded on compact $\text{SO}(3)$, and positive kinetic energy makes each energy sublevel compact.

Theorem 1 (regular reduction). *A nonsteady regular inclination has finite period if and only if its allowed component of $P \geq 0$ is bounded by two simple roots in $(-1, 1)$. In the pole-incompatible chamber $L \neq \pm G$, label the roots*

$$-1 < r_1 < r_2 < 1 < r_3, \quad P = 2A\gamma(u - r_1)(u - r_2)(u - r_3).$$

This is the complete root classification for the regular reduced motion. The formulas below give its least reduced period, reconstruction, and full closure test.

Proof of the reduction. Allowed positions are exactly $P \geq 0$. At a simple turning root the time integral is locally $\int s^{-1/2} ds$, while a double root gives $\int s^{-1} ds$. Equation (3), the positive leading coefficient, and the cubic sign pattern give the stated ordering. This also separates finite nutation from a separatrix.

2 Elliptic inversion and two phase increments

Set

$$d = r_2 - r_1, \quad R = r_3 - r_1, \quad k^2 = \frac{d}{R}, \quad \nu^2 = \frac{\gamma R}{2A}, \quad Q_0 = \sqrt{2A\gamma R}.$$

Direct substitution proves

$$u(t) = r_1 + d \operatorname{sn}^2(\nu(t - t_0), k), \quad T = \frac{4A}{Q_0} K(k). \quad (4)$$

Indeed $z = (u - r_1)/d$ obeys $A^2 d^2 \dot{z}^2 = 2A\gamma d^2 R z(1 - z)(1 - k^2 z)$.

Hamilton's equations are

$$\dot{\phi} = \frac{L - Gu}{A(1 - u^2)}, \quad \dot{\psi} = \frac{G}{C} - u\dot{\phi}.$$

Define the finite complete integrals

$$I_N = \frac{2}{Q_0(1 - r_1)} \Pi\left(\frac{d}{1 - r_1}, k\right), \quad I_S = \frac{2}{Q_0(1 + r_1)} \Pi\left(-\frac{d}{1 + r_1}, k\right).$$

Partial fractions on the outbound and return legs give the two phase reconstruction

$$\Delta\phi = (L - G)I_N + (L + G)I_S, \quad (5)$$

$$\Delta\psi = G(1/C - 1/A)T + (G - L)I_N + (G + L)I_S. \quad (6)$$

Theorem 2 (full regular closure). *The full phase state in $T^*\text{SO}(3)$ closes if and only if*

$$\frac{\Delta\phi}{2\pi} \in \mathbb{Q}, \quad \frac{\Delta\psi}{2\pi} \in \mathbb{Q}.$$

If their reduced denominators are q_ϕ, q_ψ , the least number of nutations in a full return is $\text{lcm}(q_\phi, q_\psi)$.

Proof. After one least return of $(u, \dot{u})$, autonomy makes the two angle increments constant. Rational increments therefore suffice. Conversely, a full return must occur after an integer number of reduced returns. For $0 < \theta < \pi$, the z - y - z decomposition is unique modulo independent 2π shifts of ϕ, ψ , so both increments must be rational multiples of 2π . $\square$